# Patent Prior-Art & Litigation Analytics Engine

*AI-Based Patent Search, Claim Similarity, Citation Analysis & Prior-Art Optimisation*

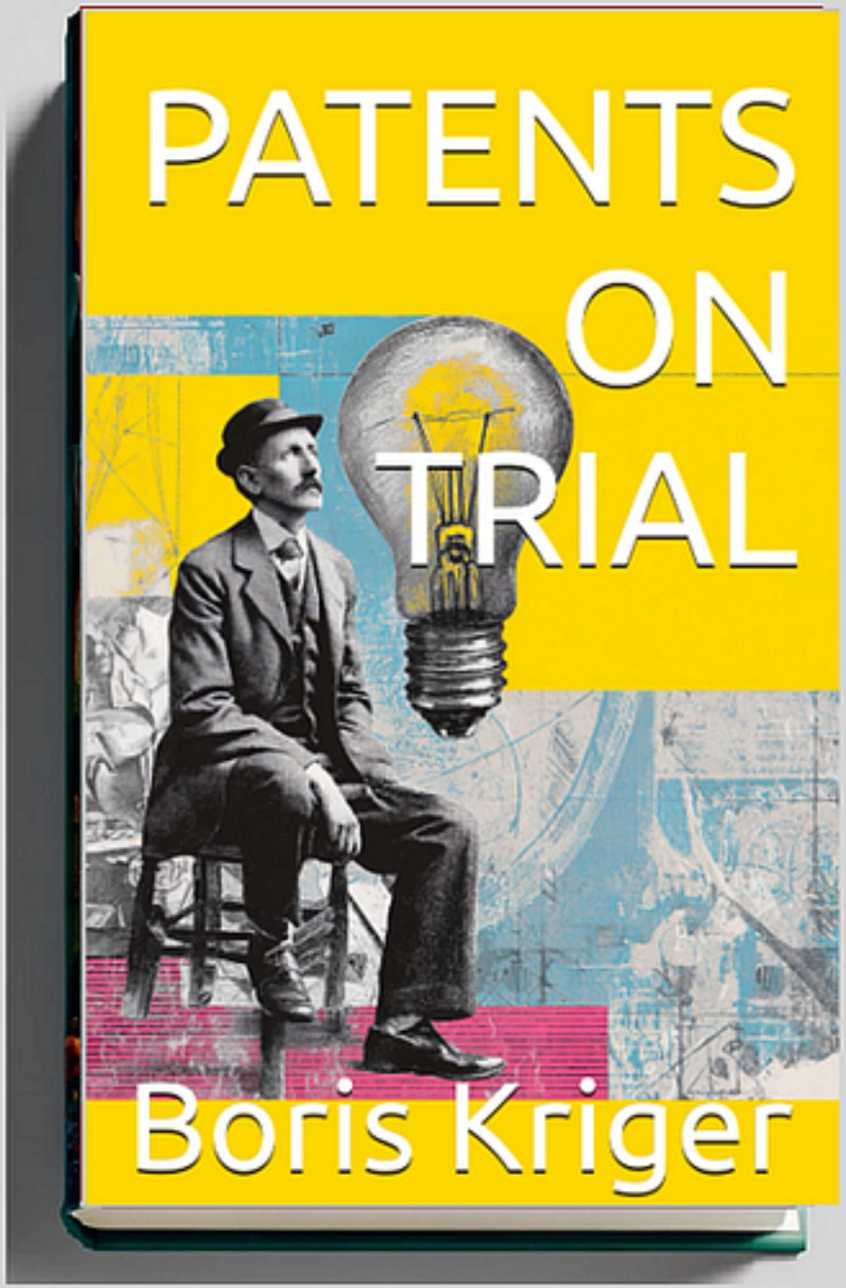
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ACADEMIC PROJECT PRESENTATION




PROBLEM STATEMENT

# The Challenge in Patent Examination

### Scale & Complexity

Millions of patent documents make manual prior-art discovery impractical.

### Semantic Gap

Similar inventions use different terminology, causing keyword search to miss relevant prior art.

### Citation Networks

Complex citation relationships are difficult to analyze manually.

### Reference Optimization

Identifying the smallest prior-art combination covering all claim features is computationally challenging.

**Core Problem:** Develop an intelligent analytics engine that searches patent claims, identifies semantically similar prior art, analyzes citation influence, computes a minimum prior-art reference set, and recommends examiner assignment.

SOLUTION OVERVIEW

# Objectives & Proposed Solution

01

Intelligent Patent Search

02

Claim Extraction & Comparison

03

Semantic Similarity Measurement

04

Prior-Art Ranking

05

Citation Network Analysis

06

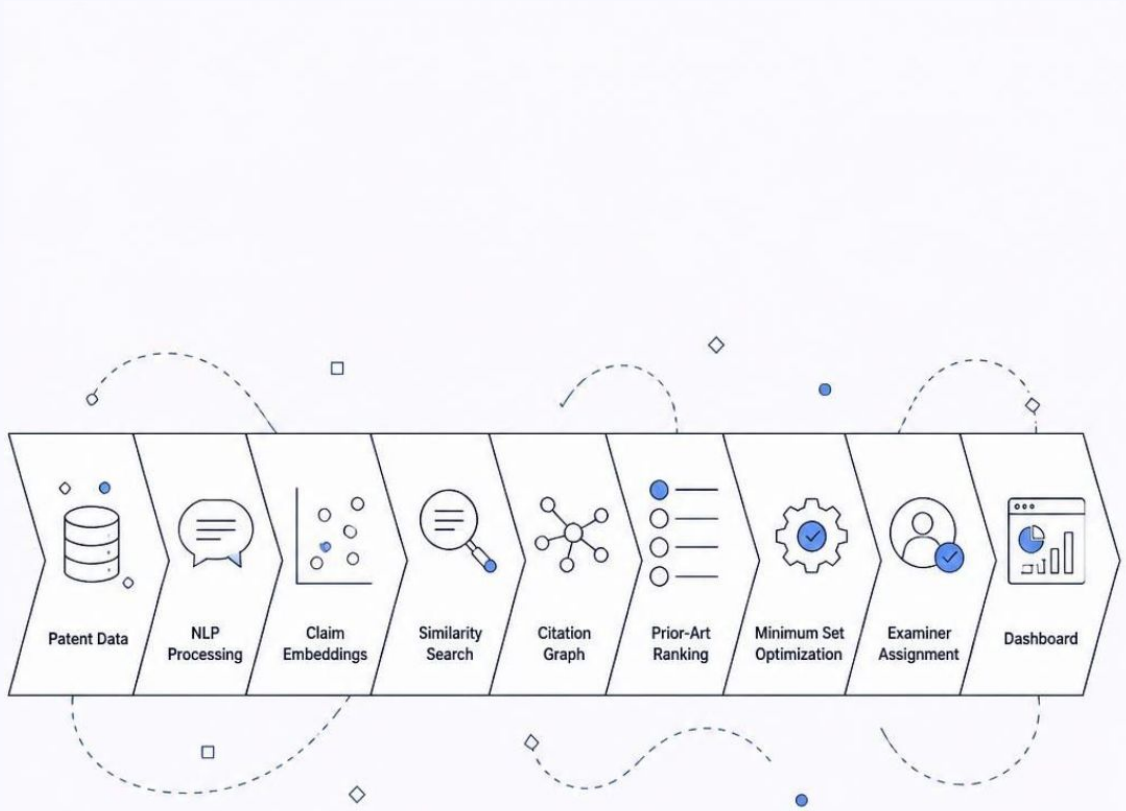
Minimum Set Optimization

07

Examiner Assignment

08

Analytics Dashboard



TECHNICAL STACK

# Software & Hardware Requirements

## Software Stack

| | |
|---|---|
| **Language** | Java |
| **Frontend** | React.js / HTML / CSS / JS |
| **Backend** | FastAPI / Flask |
| **Database** | PostgreSQL / MongoDB |
| **NLP / ML** | Sentence Transformers, spaCy, scikit-learn |
| **Graph Analysis** | NetworkX |
| **Visualization** | Plotly / D3.js / Chart.js |
| **Optimization** | Greedy Set Cover / ILP |
| **Deployment** | Docker |

## Hardware Requirements

### Minimum

Intel Core i5 / Ryzen 5 · 8 GB RAM · 256 GB SSD · Internet

### Recommended

Intel Core i7 / Ryzen 7 · 16–32 GB RAM · 512 GB–1 TB SSD · GPU for NLP

RESEARCH BACKGROUND

# Literature Survey & Existing Approaches

## Patent Information Retrieval

Boolean search, TF-IDF, IPC/CPC classification, inverted indexes.
**Limitation:** keyword mismatch.

## Patent Claim Similarity

Word/sentence embeddings, transformer models, cosine similarity for semantic claim comparison.

## Citation Network Analysis

Directed citation graphs, PageRank, degree and betweenness centrality, community detection.

## Prior-Art Optimization

Set Cover Problem, Greedy Set Cover, Integer Linear Programming for weighted reference selection.

**Research Gap:** Existing approaches address search, similarity, citation analysis, or optimization separately. The proposed system integrates all into one decision-support platform.

METHODOLOGY

# Proposed Methodology




## Key Processing Steps

- Patent claims are extracted and normalized from raw documents
- Claims are converted into TF-IDF vectors and/or semantic embeddings
- Cosine similarity ranks relevant prior-art patents
- Citation relationships are represented as a directed graph
- Set-cover optimization identifies a minimal reference set covering all claim features
- Examiner assignment considers technical expertise and workload balance

AI CORE

# AI & Algorithmic Techniques

## NLP & Semantic Search

Sentence Transformers · Embeddings · Cosine Similarity

## Information Retrieval

TF-IDF · Keyword Matching · IPC/CPC Filtering

## Citation Analytics

PageRank · Degree Centrality · Betweenness Centrality

## Optimization

Set Cover · Greedy Algorithm · Integer Linear Programming

## Examiner Assignment

Domain Matching · Expertise Profiling · Workload Balancing

## Cosine Similarity

$$\text{Similarity}(A, B) = \frac{A \quad B}{\|A\| \times \|B\|}$$

## Set-Cover Example

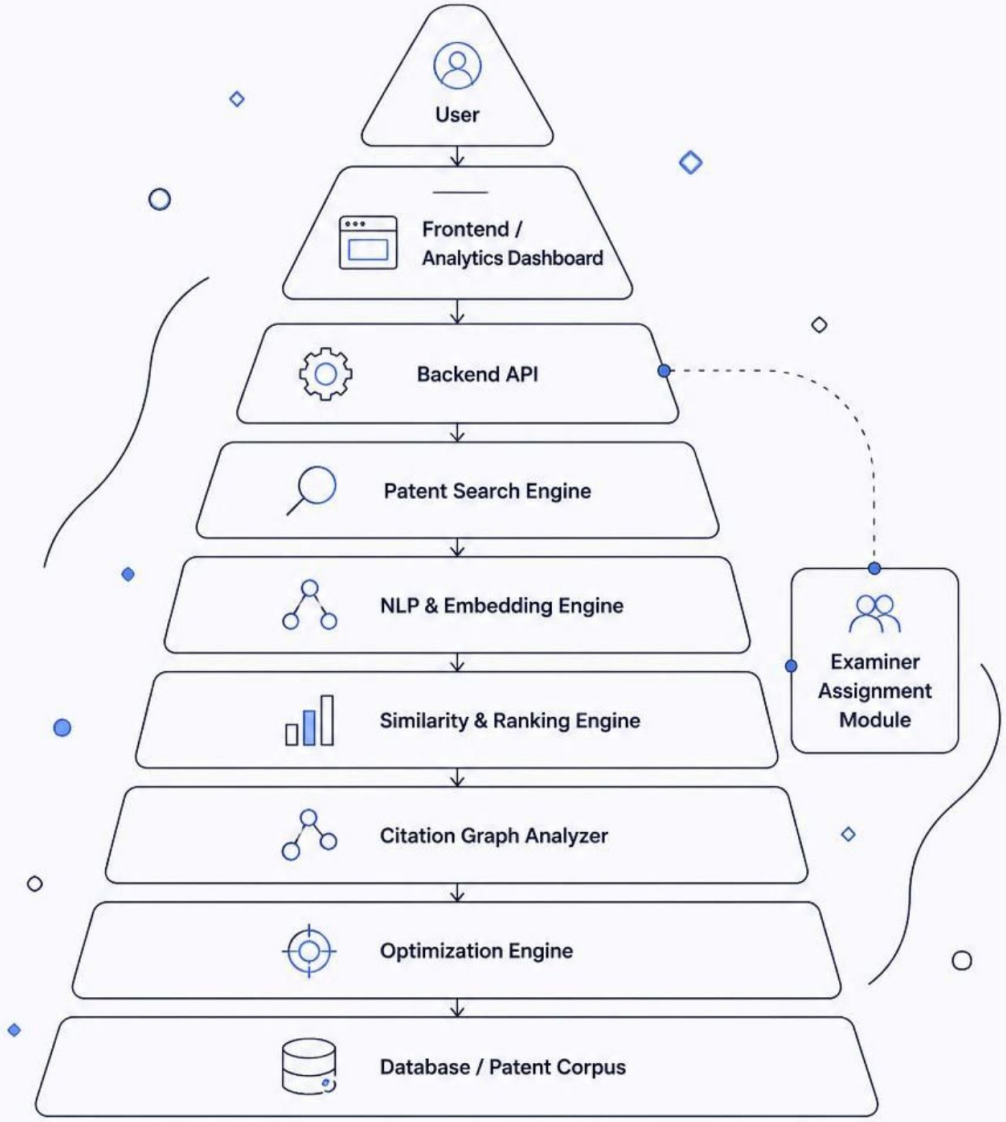
**Target Claim:** Features A + B + C + D

- P1 → covers A, B
- P2 → covers C
- P3 → covers D

**Minimum Set = {P1, P2, P3}**

ARCHITECTURE

# System Architecture




## Working Flow

- User submits target patent or application
- System extracts claims and searches the patent corpus
- NLP models calculate similarity and rank prior art
- Citation graph and claim-feature coverage are analyzed
- System recommends minimum prior-art references and examiner assignment

OUTCOMES

# Survey & Expected Outcomes

## Survey Focus Areas

- Difficulty of current patent searching
- Effectiveness of keyword-based search
- Need for semantic claim comparison
- Usefulness of citation visualization
- Value of automatic prior-art ranking
- Minimum reference-set identification
- Examiner workload and skill matching

## Expected Results

### Faster Discovery

Accelerated prior-art retrieval

### Better Comparison

Semantic claim matching

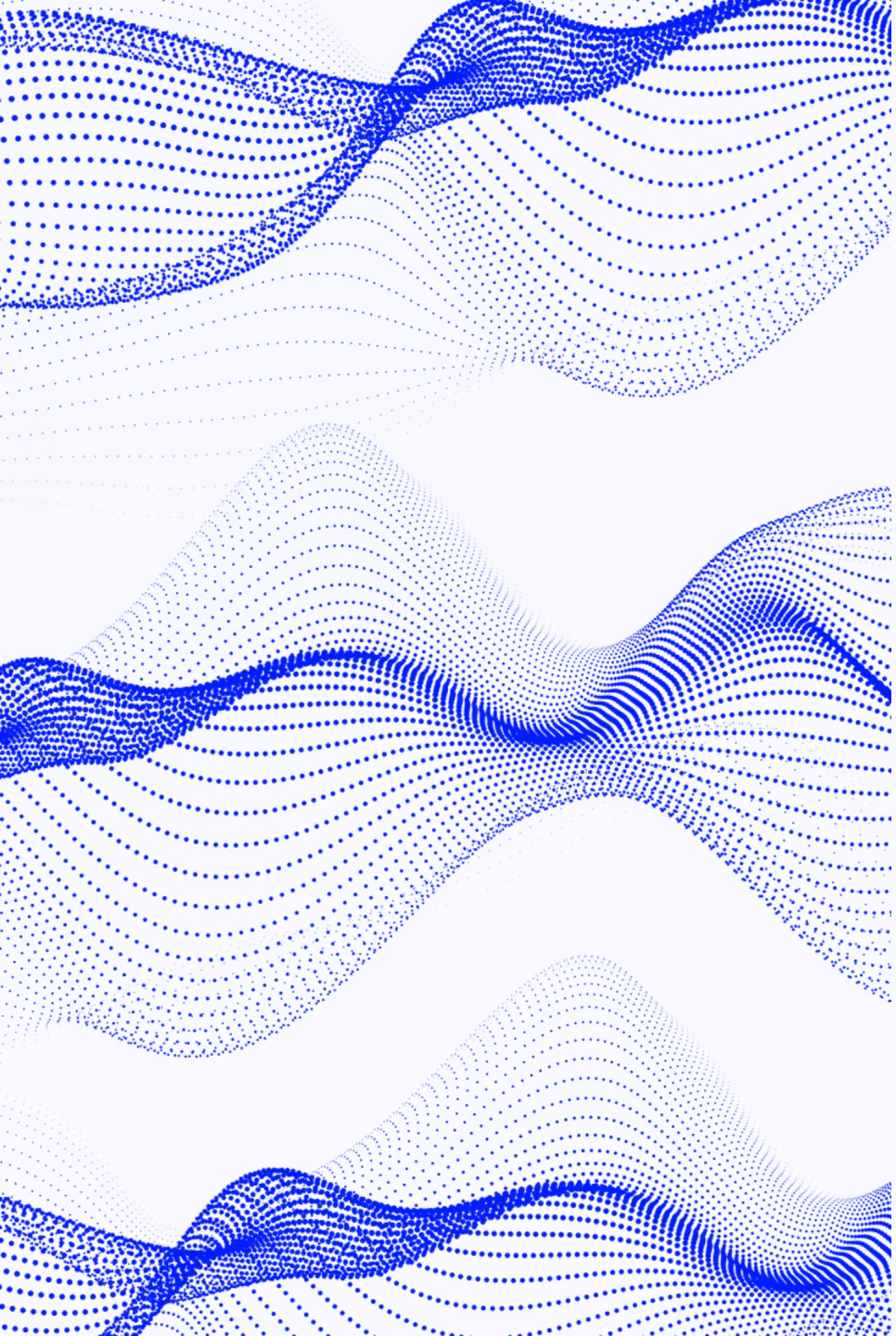
### Deeper Analysis

Improved citation insights

### Smarter Decisions

Ranked evidence for examination

**Benefits:** Search Faster → Compare Better → Analyze Deeper → Decide Smarter



CONCLUSION

# Conclusion & Future Scope

## What We Built

An integrated decision-support platform combining patent retrieval, NLP-based claim similarity, citation network analysis, prior-art ranking, set-cover optimization, and examiner assignment.

## Future Scope

- Integration with real-world patent databases
- Fine-tuned legal NLP transformer models
- Multilingual patent analysis
- Explainable AI for similarity scores
- Automated claim-chart generation
- Litigation outcome prediction
- Real-time patent monitoring
- Cloud-scale distributed processing

> **From Patent Search to Intelligent Legal Analytics**

**Thank You — Questions?**